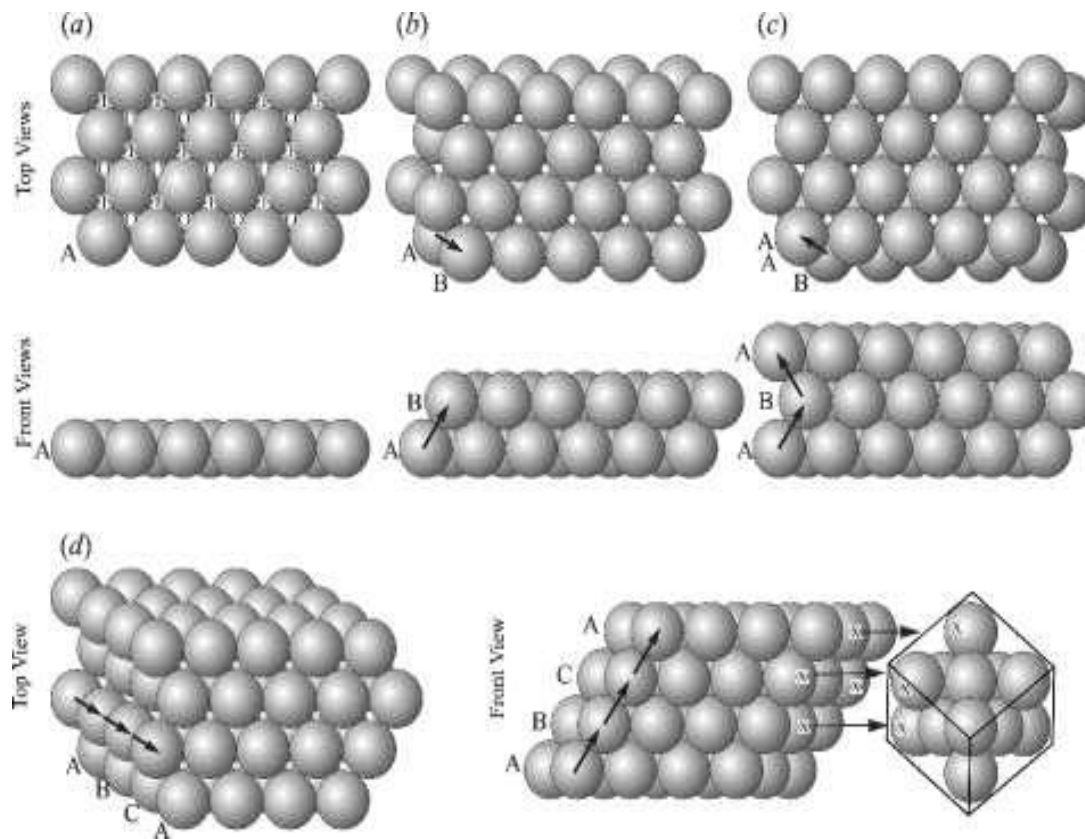


- The crystal Structures of Ionic Materials:
Pauling groups, Crystal types: simple cubic (c), body centered cubic (bcc), face centered cubic (fcc/ccp) hexagonal close packed (hcp) etc., tetrahedral and octahedral holes, coordination number, unit cell, crystal systems, Miller indices and their significance.

- We can assume minerals to be made of orderly packing of atoms or rather ions or molecules.
- Many mineral properties like symmetry, density etc are dependent on how the atoms or ions are packed
- We can approximate the atoms or ions in a metal to be rigid spherical balls

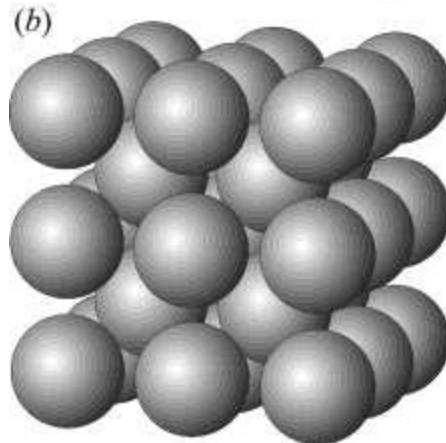
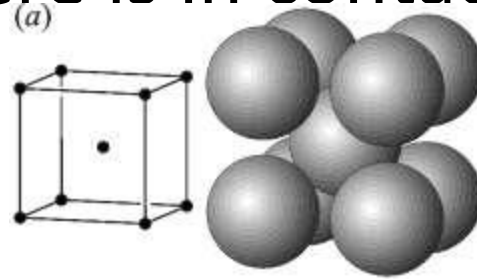
- Think of a box full of ping pong balls.
- The densest packing is one on which a layer of balls sit on the hollow between the balls in the layer below.
- The two closest packing are:
 - Hexagonal close packing (ABCABC)
 - Cubic close packing (ABAB)



Closest packing of spheres.

- Some what less dense packing is body centered cubic packing, seen in Iron.
- Because they can be packed so closely, metals have high density

- Body-centered cubic packing.
- (a) Bodycentered lattice (left) and with atoms at each lattice node (right).
- (b) Eight unit cells.
- Each sphere is in contact with eight neighbors.



Covalent and Molecular Bonds

- Covalent bonds are strong therefore covalent crystals are strong and have high melting point
- Because atoms need to be placed in a specific orientation only, covalent crystals do not show a close packing.
- The bonds in molecular crystals are weak, hence these tend to be soft.
- The geometry of the structure is determined by the shape and charge distribution in the molecules (e.g., Hydrogen bond in water)

Ionic Bonds

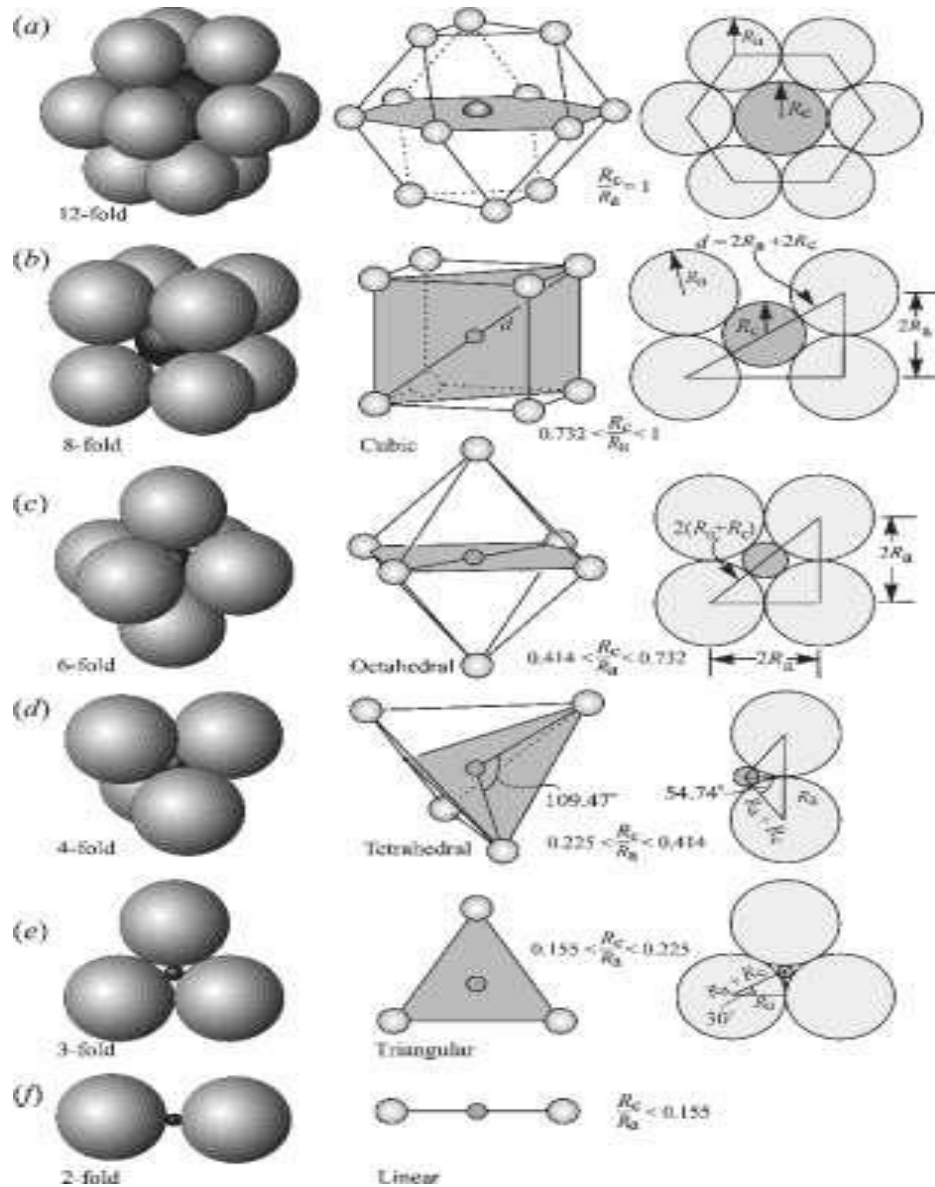
- For minerals formed largely by ionic bonding, the ion geometry can be simply considered to be spherical
- Spherical ions will geometrically pack (coordinate) oppositely charged ions around them as tightly as possible while maintaining charge neutrality
- For a particular ion, the surrounding coordination ions define the apices of a polyhedron
- The number of surrounding ions is the Coordination Number

Pauling's Rules of Mineral Structure

- **Rule 1:**
- **A coordination polyhedron** of anions is formed around each cation, wherein:
 - - the cation-anion distance is determined by the sum of the ionic radii, and
 - - the coordination number of the polyhedron is determined by the cation/anion radius ratio ($R_c:R_x$)

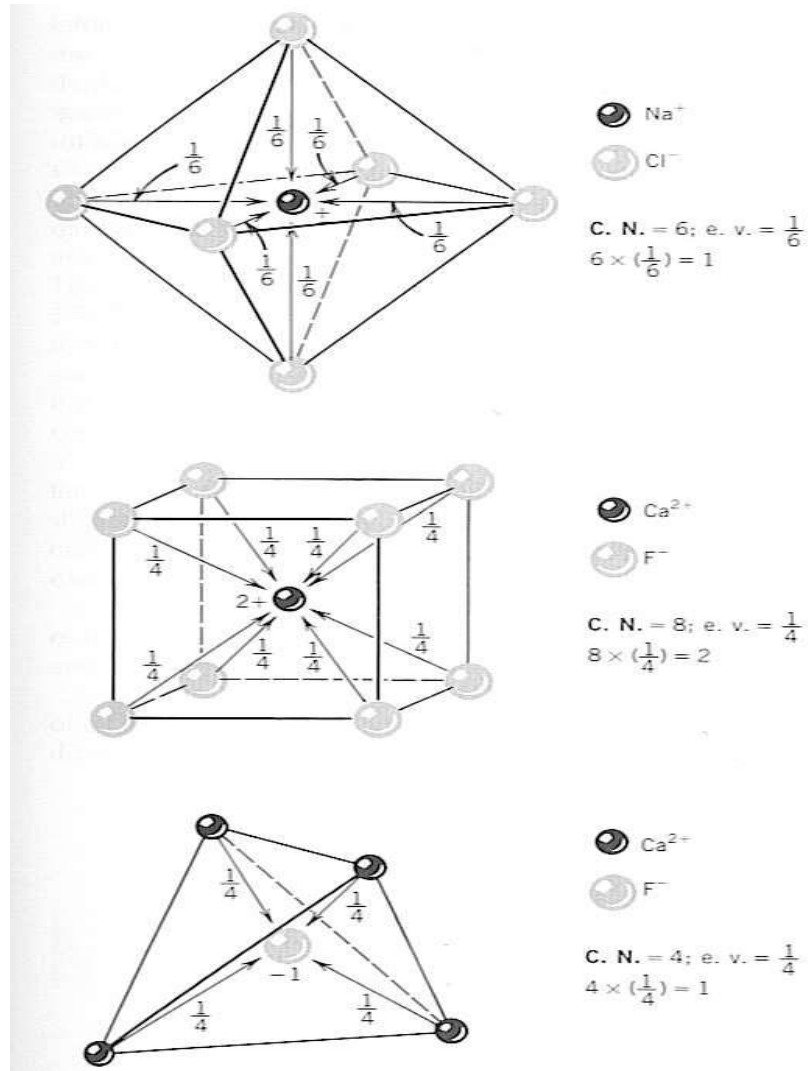
- Radius Ratio of Cation and Anion (R_c/R_a), Coordination number and Coordination polyhedra

Minimum Radius Ratio	Coordination number	Packing geometry
~1.0	12	Corners of a cuboctahedron (close packing)
0.732-1	8	Corners of a cube
0.414-0.732	6	Corners of an octahedron
0.225-0.414	4	Corners of a tetrahedron
0.155-0.225	3	Corners of an equilateral triangle
<0.155	2	Linear



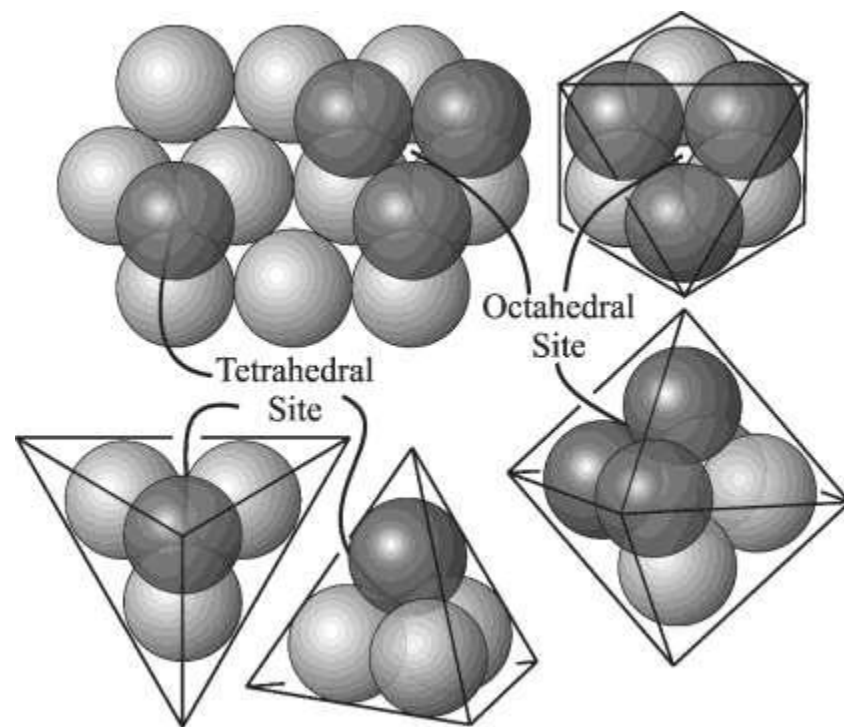
- **Rule 2: The electrostatic valency principle**
- The strength of an ionic (electrostatic) bond (e.v.) between a cation and an anion is equal to the charge of the anion (z) divided by its coordination number (n):
- $e.v. = z/n$
- In a stable (neutral) structure, a charge balance results between the cation and its polyhedral anions with which it is bonded.

Charge Balance of Ionic Bonds



Uniform Bond Strength

- **Isodesmic: where all ionic** bonds are of equal strength
- Anions tend to pack around the cation in highly symmetric and generally close packing arrangement
- Most of the minerals crystallize in highly symmetric systems like isometric, tetragonal or hexagonal systems
- Oxides, Fluorides, Chlorides are common examples



Nonuniform Bond Strength

- Some anion-cation bonds are stronger than others: cations forming stronger bonds are smaller and have a high charge
 - – Eg: C^{4+} , S^{6+} , P^{5+} , Si^{4+}
 - – These strongly charged cations form CO_3^{2-} , SO_4^{4-} , PO_4^{3-} , SiO_4^{4-} anionic groups
- Have lower symmetry, irregular coordination polyhedra

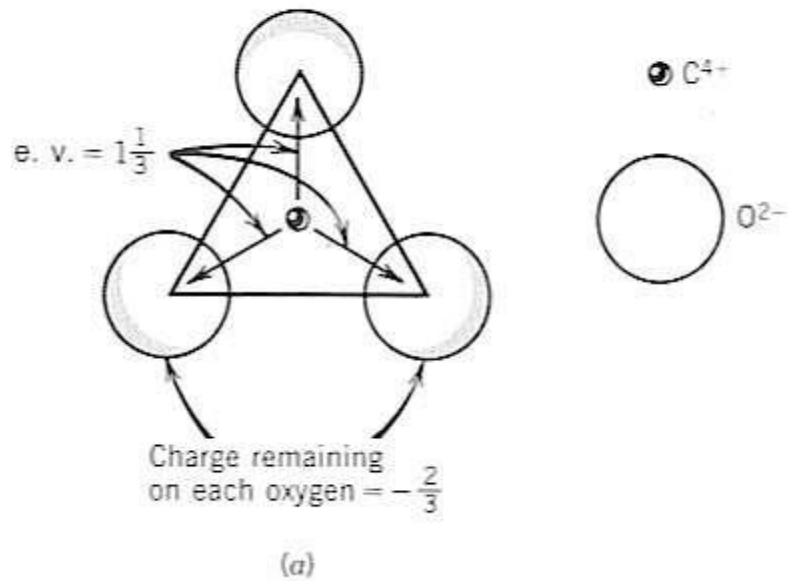
Formation of Anionic Groups

- **Anisodesmic Bonds results from high valence cations with electrostatic** valencies greater than half the valency of the polyhedral anions; other bonds with those anions will be relatively weaker.
- Mesodesmic: anion-cation bond takes exactly half of the available anion-cation charge

Anisodesmic

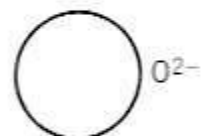
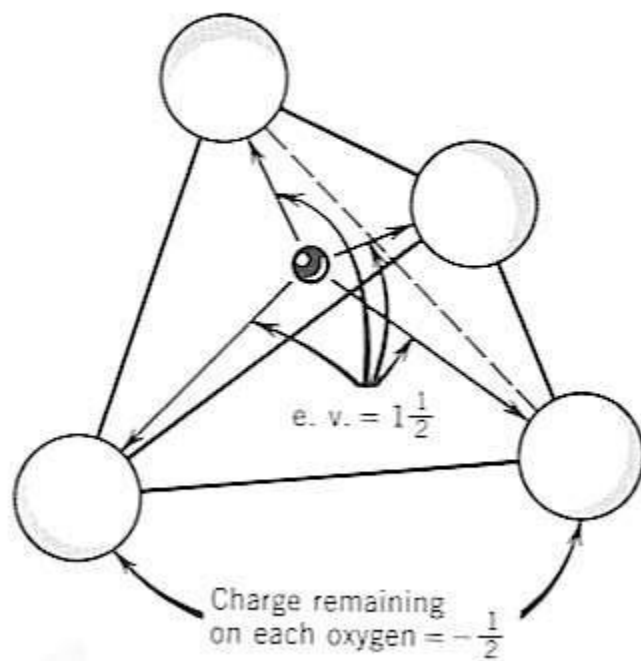
Triangular $(\text{CO}_3)^{2-}$ group

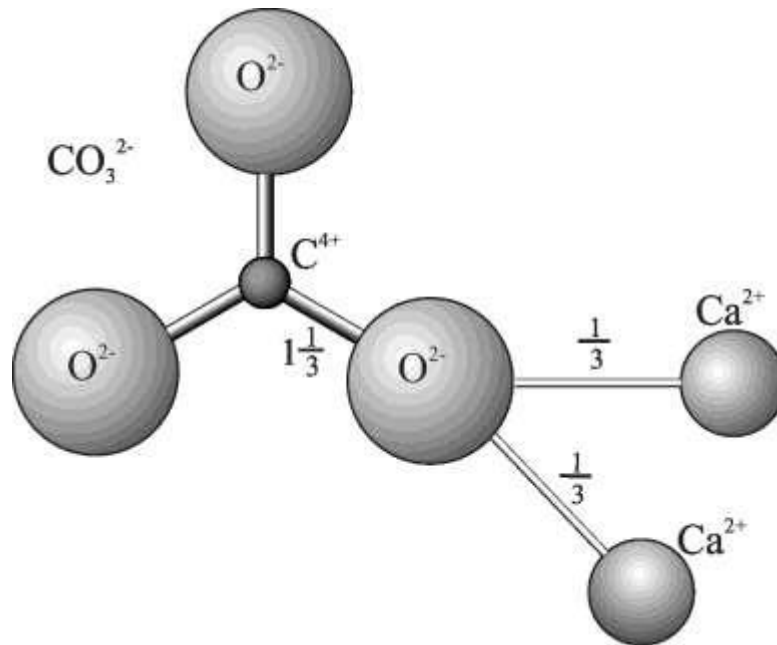
C. N. = 3; e. v. = $\frac{4}{3} = 1\frac{1}{3}$



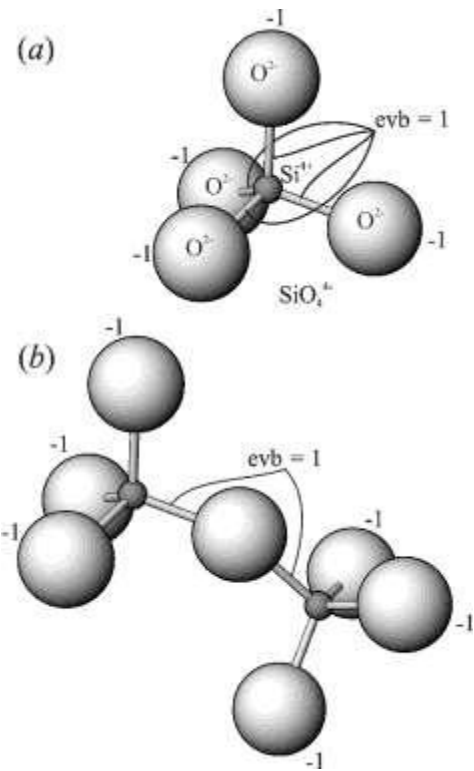
Tetrahedral $(\text{SO}_4)^{2-}$ group

C. N. = 4; e. v. = $\frac{6}{4} = 1\frac{1}{2}$





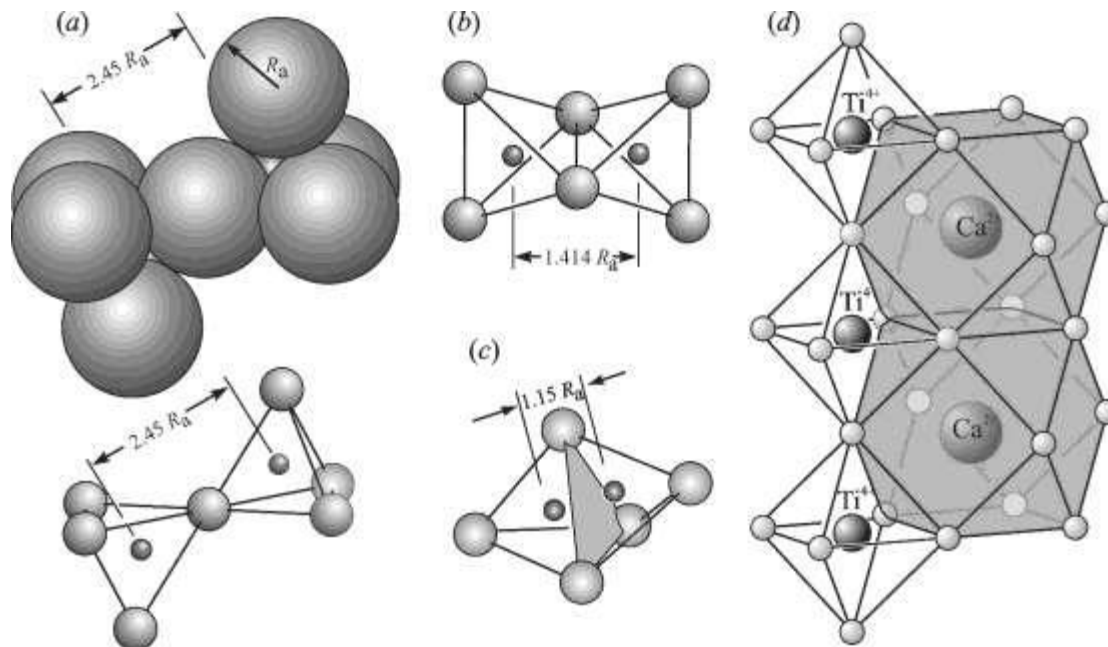
Anisodesmic carbonate (CO_3^{2-}) group. Each O–C bond occupies 1.33 of the available -2 charge on oxygen; only two-thirds of a charge on the oxygen is available to bond with other cations such as Ca^{2+} .



Mesodesmic silicon tetrahedra.

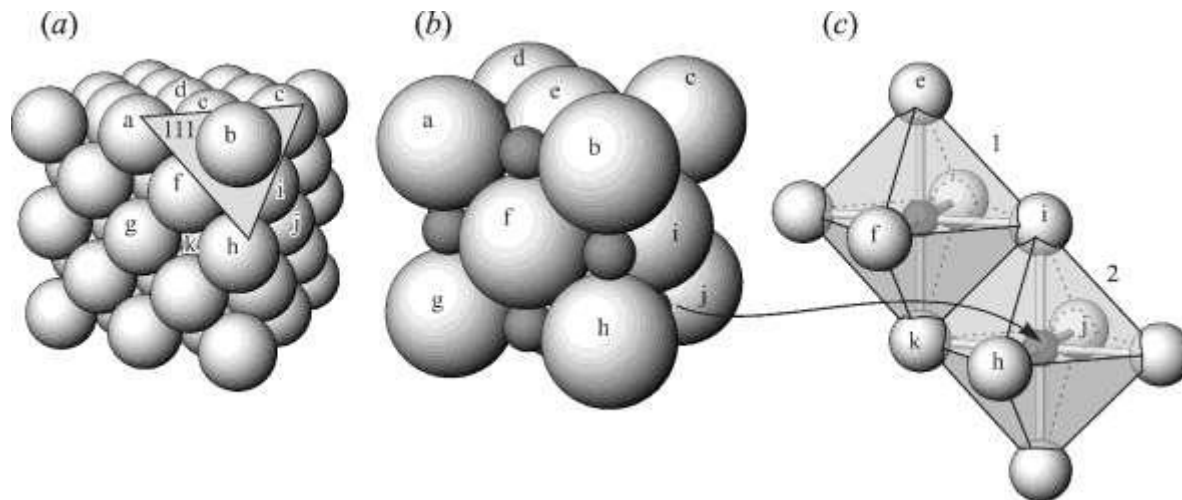
- **Rule 3: Anion polyhedra that share edges or faces decrease their stability due to bringing cations closer together; especially significant for high valency cations**

- **Rule 4: In structures with different types of cations, those cations with high valency and small CN tend not to share polyhedra with each other; when they do, polyhedra are deformed to accommodate cation repulsion**

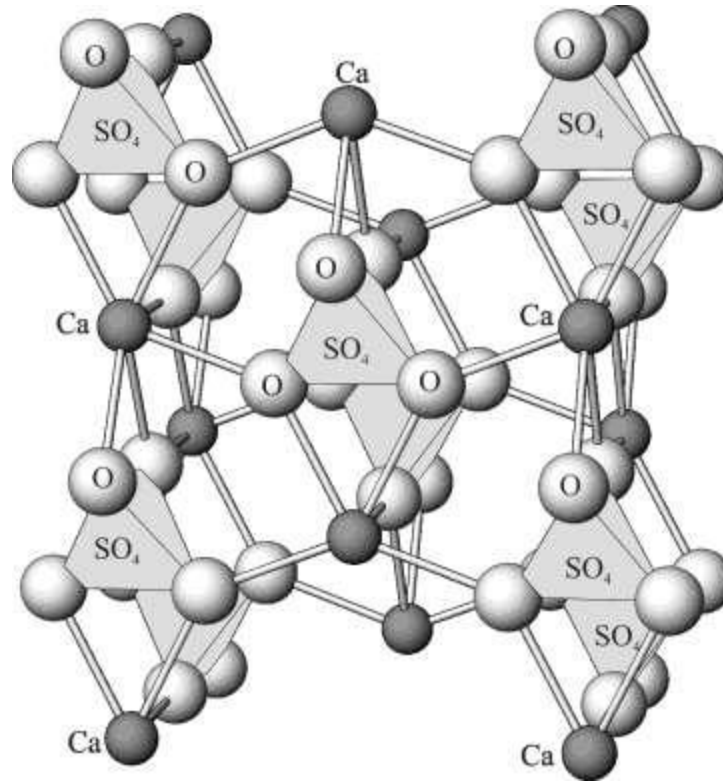


Sharing anions between polyhedra.

- **Rule 5: The principle of parsimony : nature tends towards simplicity**
- Because the number and types of different structural sites tends to be limited, even in complex minerals, different ionic elements are forced to occupy the same structural positions – leads to solid solution.



NaCl structure.



Anhydrite structure. Isolated SO_4^{2-} groups are bonded laterally through Ca^{2+} (dark spheres).

- Crystal types: simple cubic (c), body centered cubic (bcc), face centered cubic (fcc/ccp) hexagonal close packed (hcp) etc.,

SUMMARY: BONDING

Type	Bond Energy	Comments
Ionic	Large!	Nondirectional (ceramics)
Covalent	Variable large-Diamond small-Bismuth	Directional semiconductors, ceramics polymer chains)
Metallic	Variable large-Tungsten small-Mercury	Nondirectional (metals)
Secondary	smallest	Directional inter-chain (polymer) inter-molecular

SUMMARY: PRIMARY BONDS

Ceramics

(Ionic & covalent bonding):

Large bond energy

large T_m

large E

small α

Metals

(Metallic bonding):

Variable bond energy

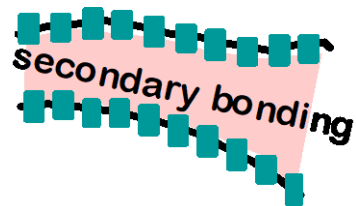
moderate T_m

moderate E

moderate α

Polymers

(Covalent & Secondary):



Directional Properties

Secondary bonding dominates

small T

small E

large α

Lattice and Unit Cells

- Lattices:
- The simplest regular array is a line of evenly spaced objects.
- There is a dot at the same place in each object.



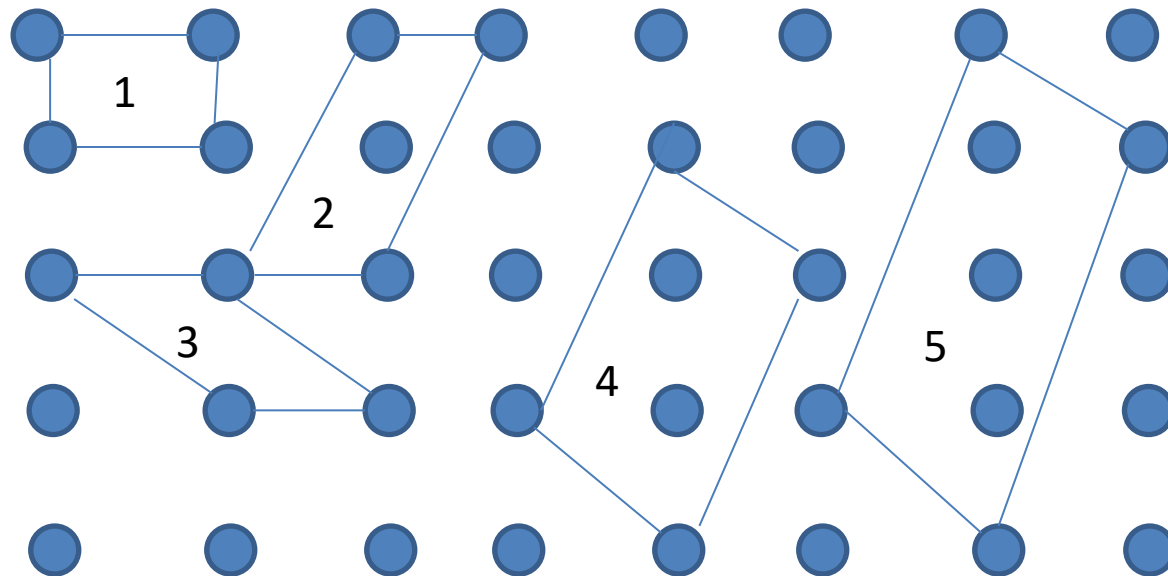
- The line of the dots is called the lattice and each lattice point must have identical surroundings.

One and Two dimensional Unit Cell

- The unit cell for the 1-D lattice lies between two vertical lines.



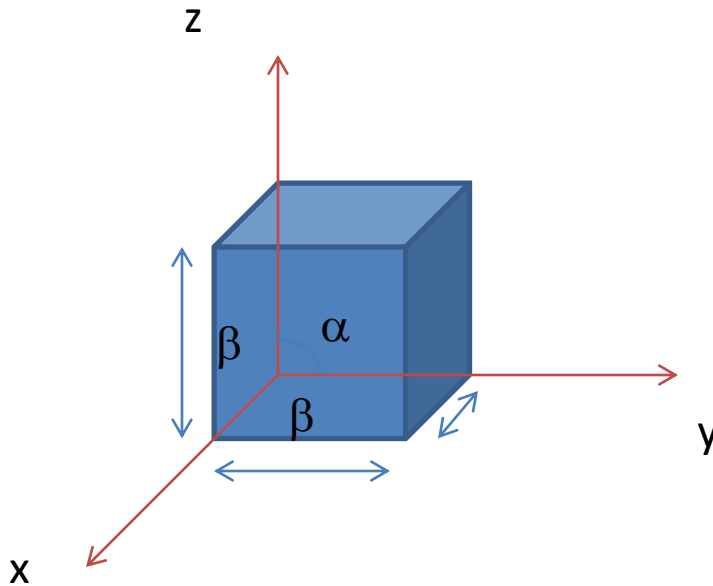
- The unit or the 2-D lattices are parallelograms with their corners at equivalent positions in the array.



- Five 2-D lattices are possible.

Three dimensional unit cell

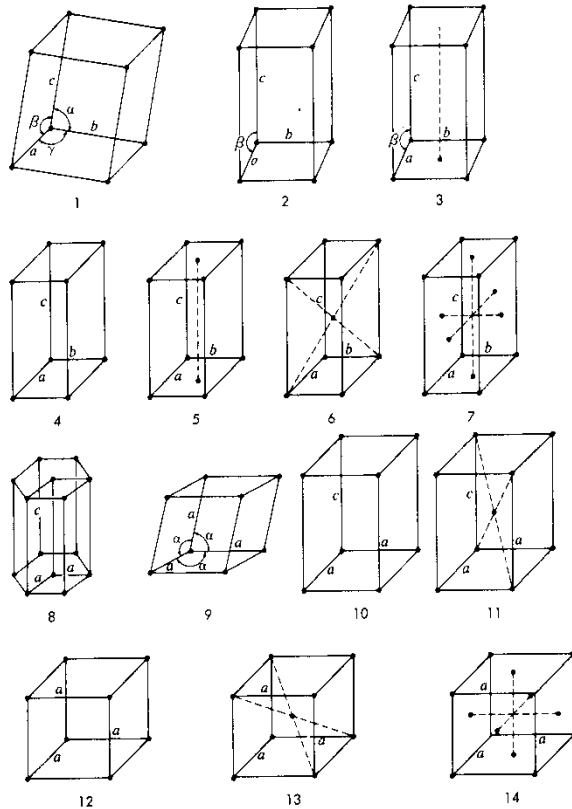
- The unit cell of a 3-D is defined by three distance a, b and c and three angles α, β and γ .



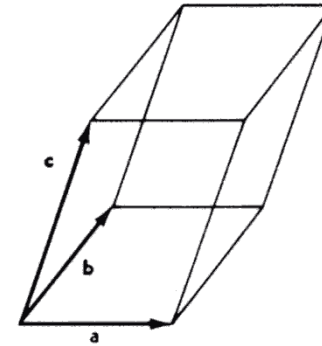
- Since the unit cells are the basic building blocks of the crystals, they must be space filling.
- The 3-D unit cell includes four different types.
- 1. The primitive cell – has a lattice point at each corner.
- 2. The body centered unit cell- has a lattice point at each corner and one at the center of the cell.
- 3. The face centered unit cell – has a lattice point at each corner and one in the center of each face.
- 4. The face centered unit cell – has a lattice point at each corner and one in the centers of one pair of opposite faces. (eg., an A centered cell has lattice point in the centers of the bc faces)

- These are known as the seven crystal systems or classes.

14 different point lattices, called *Bravais lattices*, make up the crystal system. The lengths of the sides, a , b , and c , and the angles between them can vary for a particular *unit cell*.



The 14 point lattices illustrated by a unit cell of each: (1) triclinic, simple; (2) monoclinic, simple; (3) monoclinic, base centered; (4) orthorhombic, simple; (5) orthorhombic, base centered; (6) orthorhombic, body centered; (7) orthorhombic, face centered; (8) hexagonal; (9) rhombohedral; (10) tetragonal, simple; (11) tetragonal, body centered; (12) cubic, simple; (13) cubic, body centered; (14) cubic, face centered.



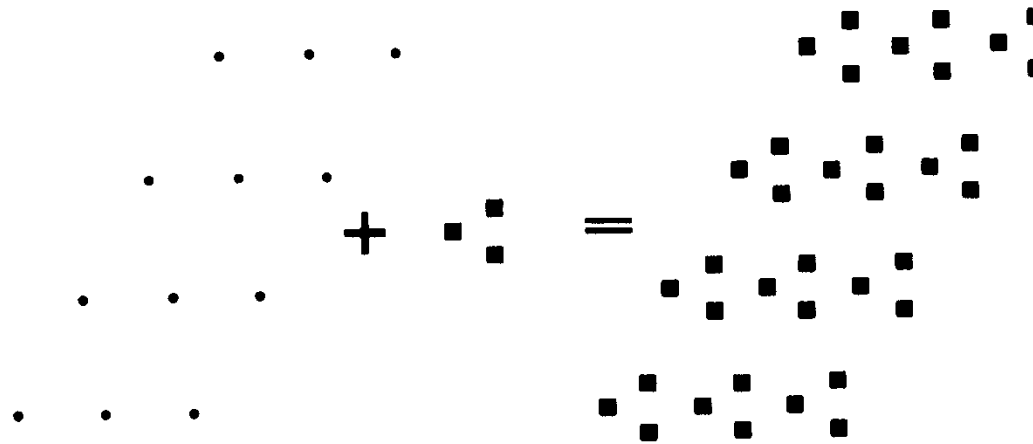
System	Axes* and Interaxial Angles	Examples
Triclinic	Three axes not at right angles, of any length $a \neq b \neq c$ $\alpha \neq \beta \neq \gamma \neq 90^\circ$	B(OH)_3 , $\text{K}_2\text{S}_2\text{O}_8$, Al_2SiO_5 , $\text{NaAlSi}_3\text{O}_8$
Monoclinic	Three axes, one pair not at right angles, of any lengths $a \neq b \neq c$ $\alpha = \gamma = 90^\circ \neq \beta$	$\text{C}_{18}\text{H}_{24}$, KNO_2 , $\text{K}_2\text{S}_4\text{O}_{16}$, As_4S_4 , KClO_3
Orthorhombic	Three axes at right angles, all unequal $a \neq b \neq c$ $\alpha = \beta = \gamma = 90^\circ$	I, Ga, Fe_3C , FeS_2 , BaSO_4
Tetragonal	Three axes at right angles, two equal $a = b \neq c$ $\alpha = \beta = \gamma = 90^\circ$	In, TiO_2 , $\text{C}_4\text{H}_{10}\text{O}_4$, KIO_4
Cubic	Three axes at right angles, all equal $a = b = c$ $\alpha = \beta = \gamma = 90^\circ$	Cu, Ag, Ar, Si, Ni, NaCl, LiF
Hexagonal	Two axes of equal length at 120° , third axis at 90° to these $a = b \neq c$ $\alpha = \beta = 90^\circ$ $\gamma = 120^\circ$	Zn, Cd, Mg, NiAs
Rhombohedral	Three axes equally inclined, not at right angles, all equal $a = b = c$ $\alpha = \beta = \gamma \neq 90^\circ$	Hg, Sb, Bi

METALLIC CRYSTALS

- tend to be densely packed.
- have several reasons for dense packing:
 - Typically, only one element is present, so all atomic radii are the same.
 - Metallic bonding is not directional.
 - Nearest neighbor distances tend to be small in order to lower bond energy.
- have the simplest crystal structures. 74 elements have the simplest crystal structures – BCC, FCC and HCP

We will look at three such structures...

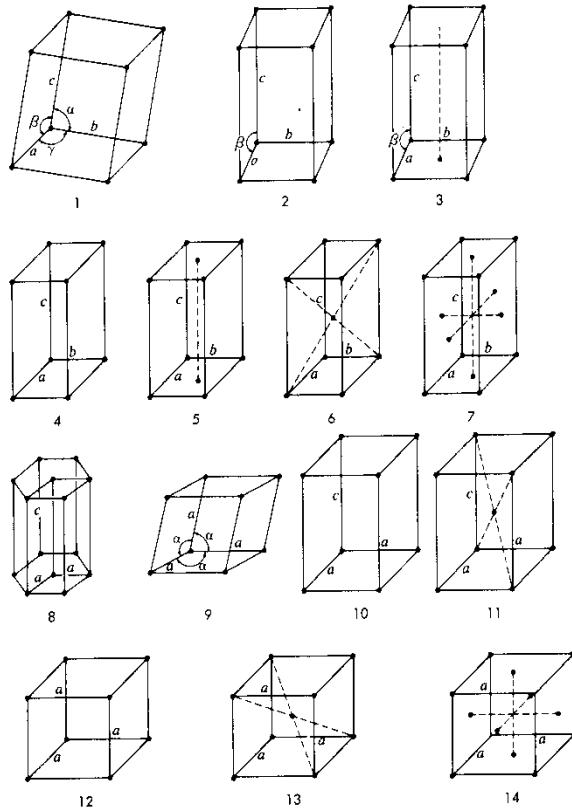
The crystal lattice



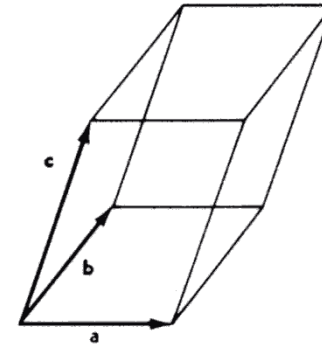
Point lattice + atom group = periodic atom array

A point lattice is made up of regular, repeating points in space. An atom or group of atoms are tied to each lattice point

14 different point lattices, called *Bravais lattices*, make up the crystal system. The lengths of the sides, a , b , and c , and the angles between them can vary for a particular *unit cell*.

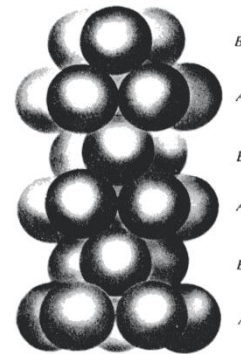
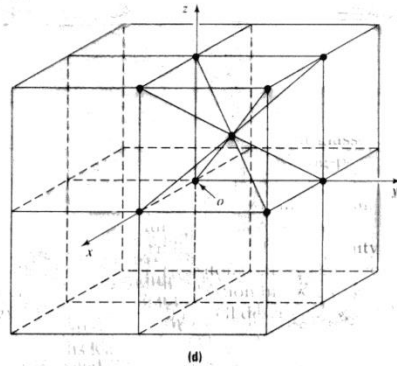
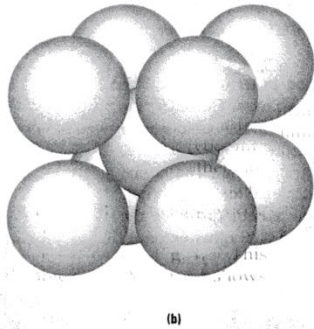
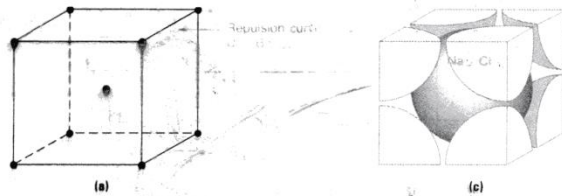


The 14 point lattices illustrated by a unit cell of each: (1) triclinic, simple; (2) monoclinic, simple; (3) monoclinic, base centered; (4) orthorhombic, simple; (5) orthorhombic, base centered; (6) orthorhombic, body centered; (7) orthorhombic, face centered; (8) hexagonal; (9) rhombohedral; (10) tetragonal, simple; (11) tetragonal, body centered; (12) cubic, simple; (13) cubic, body centered; (14) cubic, face centered.

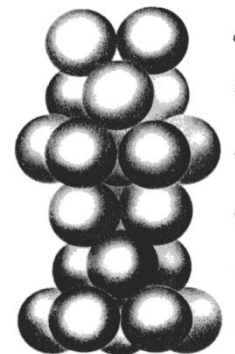


System	Axes* and Interaxial Angles	Examples
Triclinic	Three axes not at right angles, of any length $a \neq b \neq c$ $\alpha \neq \beta \neq \gamma \neq 90^\circ$	$B(OH)_3$, $K_2S_2O_8$, Al_2SiO_5 , $NaAlSi_3O_8$
Monoclinic	Three axes, one pair not at right angles, of any lengths $a \neq b \neq c$ $\alpha = \gamma = 90^\circ \neq \beta$	$C_{18}H_{24}$, KNO_2 , $K_2S_4O_6$, As_4S_4 , $KClO_3$
Orthorhombic	Three axes at right angles, all unequal $a \neq b \neq c$ $\alpha = \beta = \gamma = 90^\circ$	I, Ga, Fe_3C , FeS_2 , $BaSO_4$
Tetragonal	Three axes at right angles, two equal $a = b \neq c$ $\alpha = \beta = \gamma = 90^\circ$	In, TiO_2 , $C_4H_{10}O_4$, KIO_4
Cubic	Three axes at right angles, all equal $a = b = c$ $\alpha = \beta = \gamma = 90^\circ$	Cu, Ag, Ar, Si, Ni, NaCl, LiF
Hexagonal	Two axes of equal length at 120° , third axis at 90° to these $a = b \neq c$ $\alpha = \beta = 90^\circ$ $\gamma = 120^\circ$	Zn, Cd, Mg, NiAs
Rhombohedral	Three axes equally inclined, not at right angles, all equal $a = b = c$ $\alpha = \beta = \gamma \neq 90^\circ$	Hg, Sb, Bi

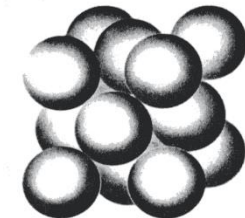
Three simple lattices that describe metals are Face Centered Cubic (FCC) Body Centered Cubic (BCC) and Hexagonal Close Packed (HCP)



(a) Hexagonal close packing

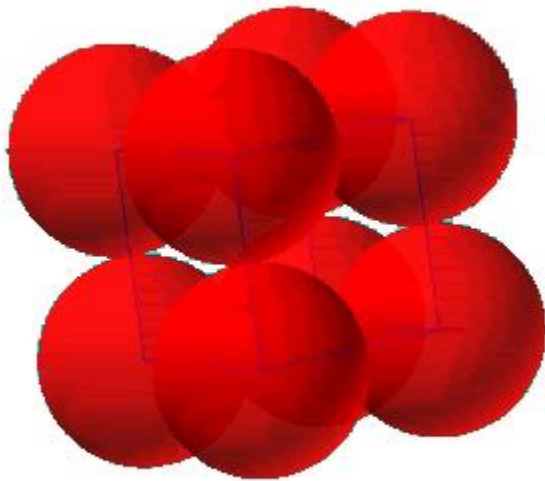


(b) Cubic close packing



SIMPLE CUBIC STRUCTURE (SC)

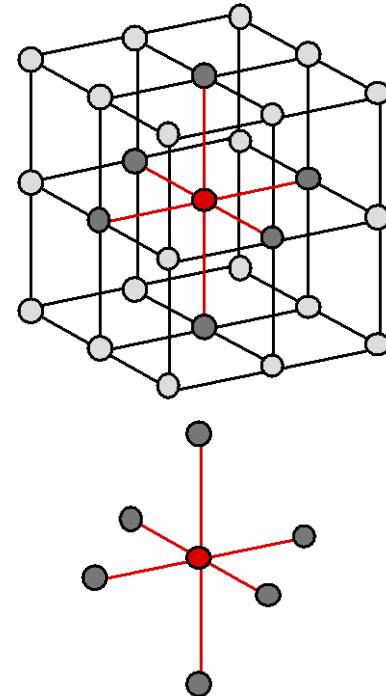
- Rare due to poor packing (only Po has this structure)
- **Close-packed directions** are cube edges.



Click on image to animate

(Courtesy P.M. Anderson)

- **Coordination # = 6**
(# nearest neighbors)

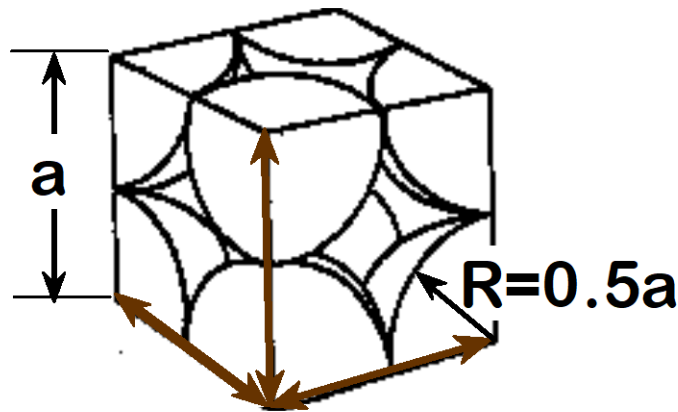


ATOMIC PACKING FACTOR

$$\text{APF} = \frac{\text{Volume of atoms in unit cell}^*}{\text{Volume of unit cell}}$$

*assume hard spheres

- APF for a simple cubic structure = 0.52



close-packed directions

contains $8 \times 1/8 =$

1 atom/unit cell

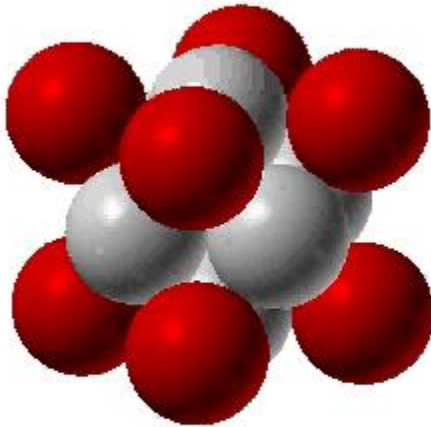
Adapted from Fig. 3.19,
Callister 6e.

$$\text{APF} = \frac{\overbrace{1}^{\text{atoms unit cell}} \overbrace{\frac{4}{3} \pi (0.5a)^3}^{\text{volume atom}}}{\underbrace{a^3}_{\text{volume unit cell}}}$$

FACE CENTERED CUBIC STRUCTURE (FCC)

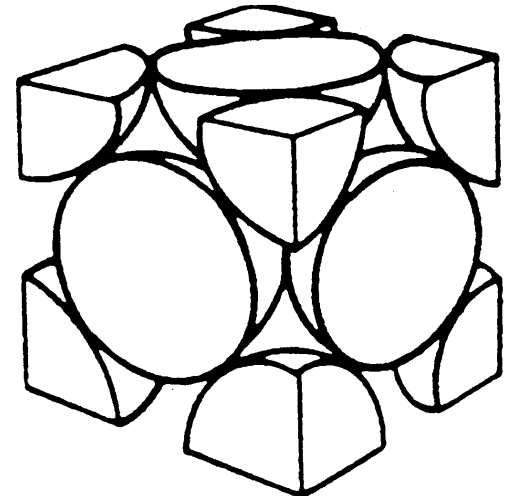
- Close packed directions are face diagonals.
--Note: All atoms are identical; the face-centered atoms are shaded differently only for ease of viewing.

- Coordination # = 12



Click on image to animate

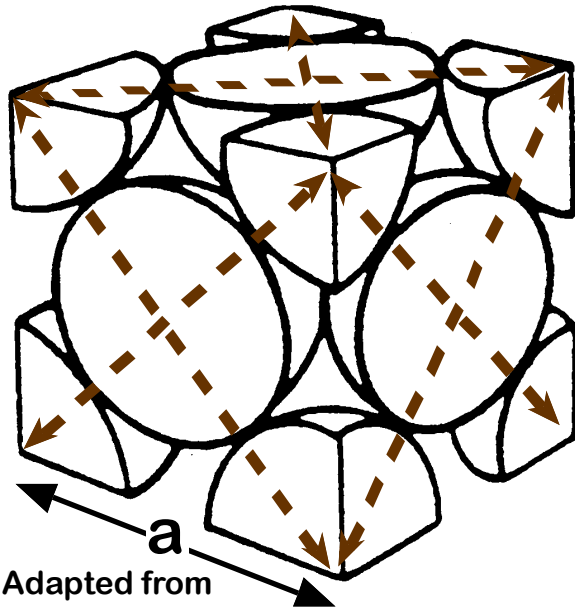
(Courtesy P.M. Anderson)



Adapted from Fig. 3.1(a),
Callister 6e.

ATOMIC PACKING FACTOR: FCC

- APF for a body-centered cubic structure = 0.74



Adapted from
Fig. 3.1(a),
Callister 6e.

Close-packed directions:
length = $4R$
 $= \sqrt{2} a$

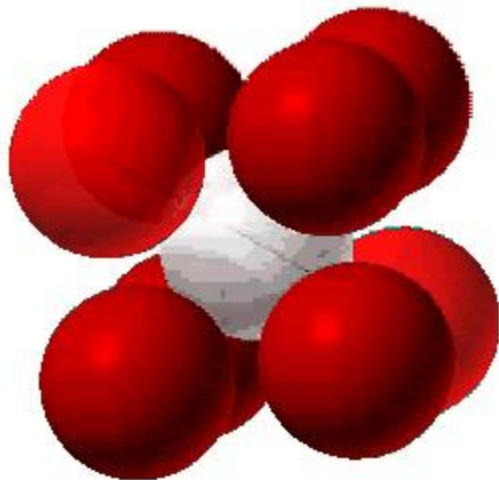
Unit cell contains:
 $6 \times 1/2 + 8 \times 1/8$
 $= 4 \text{ atoms/unit cell}$

$$\text{APF} = \frac{\text{atoms unit cell} \times \frac{4}{3} \pi (\sqrt{2}a/4)^3}{a^3}$$

$\xleftarrow{\text{volume atom}}$
 $\xleftarrow{\text{volume unit cell}}$

BODY CENTERED CUBIC STRUCTURE (BCC)

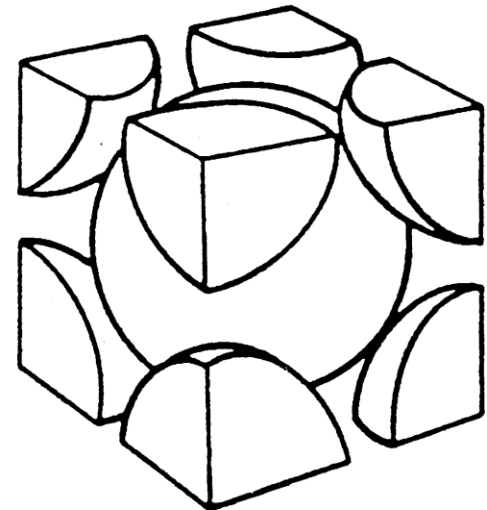
- Close packed directions are cube diagonals.
--Note: All atoms are identical; the center atom is shaded differently only for ease of viewing.



Click on image to animate

(Courtesy P.M. Anderson)

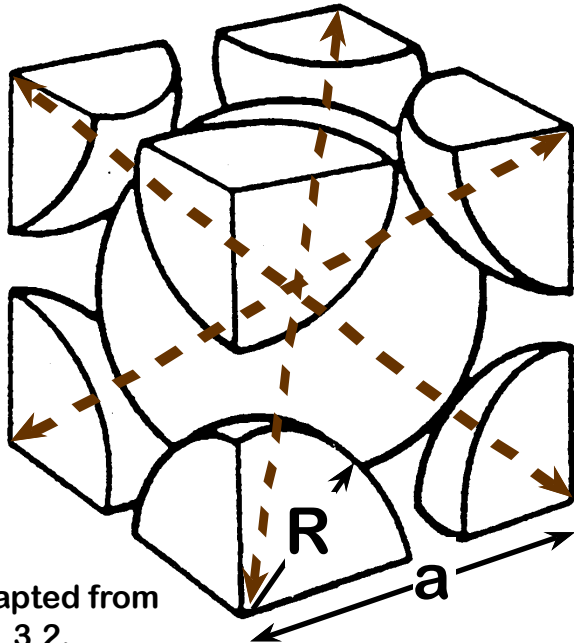
- Coordination # = 8



Adapted from Fig. 3.2,
Callister 6e.

ATOMIC PACKING FACTOR: BCC

- APF for a body-centered cubic structure = 0.68



Adapted from
Fig. 3.2,
Callister 6e.

Close-packed directions:
length = $4R$
 $= \sqrt{3} a$

Unit cell contains:
 $1 + 8 \times 1/8$
 $= 2 \text{ atoms/unit cell}$

$$\text{APF} = \frac{\text{atoms unit cell} \times \frac{4}{3} \pi \left(\frac{\sqrt{3}a}{4} \right)^3}{a^3}$$

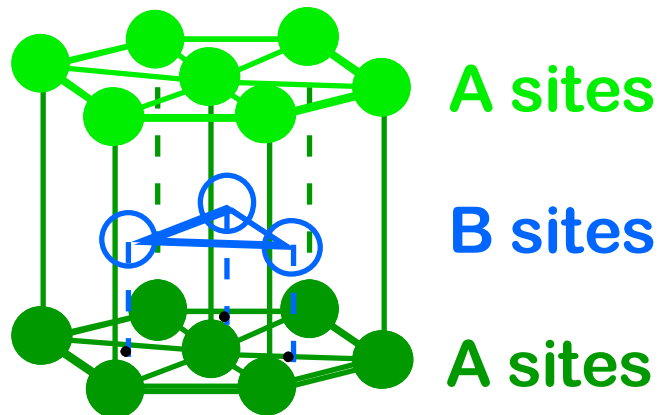
Labels for the equation components:

- atoms unit cell** (green text) points to the '2' in the numerator.
- volume atom** (brown text) points to the $\frac{4}{3} \pi \left(\frac{\sqrt{3}a}{4} \right)^3$ term in the numerator.
- volume unit cell** (blue text) points to the a^3 term in the denominator.

HEXAGONAL CLOSE-PACKED STRUCTURE (HCP)

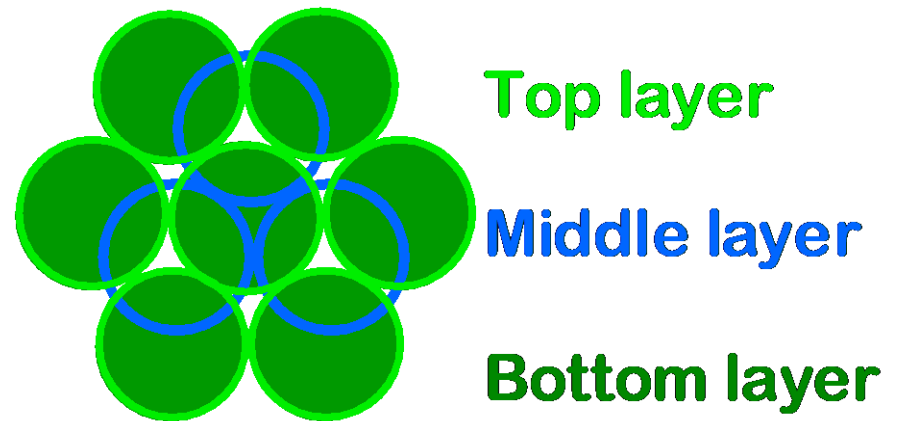
- ABAB... Stacking Sequence

- 3D Projection



Adapted from Fig. 3.3,
Callister 6e.

- 2D Projection



- Coordination # = 12

- APF = 0.74

THEORETICAL DENSITY, ρ

$$\rho = \frac{n A}{V_c N_A}$$

atoms/unit cell $\rightarrow$ n Atomic weight (g/mol) $\rightarrow$ A
 Volume/unit cell $\rightarrow$ V_c Avogadro's number $\rightarrow$ N_A
 ($\text{cm}^3/\text{unit cell}$) ($6.023 \times 10^{23} \text{ atoms/mol}$)

Example: Copper

Data from Table inside front cover of Callister (see next slide):

- crystal structure = FCC: 4 atoms/unit cell
- atomic weight = 63.55 g/mol (1 amu = 1 g/mol)
- atomic radius $R = 0.128 \text{ nm}$ (1 nm = 10^{-7} cm)

$$V_c = a^3; \text{ For FCC, } a = 4R/\sqrt{2}; V_c = 4.75 \times 10^{-23} \text{ cm}^3$$

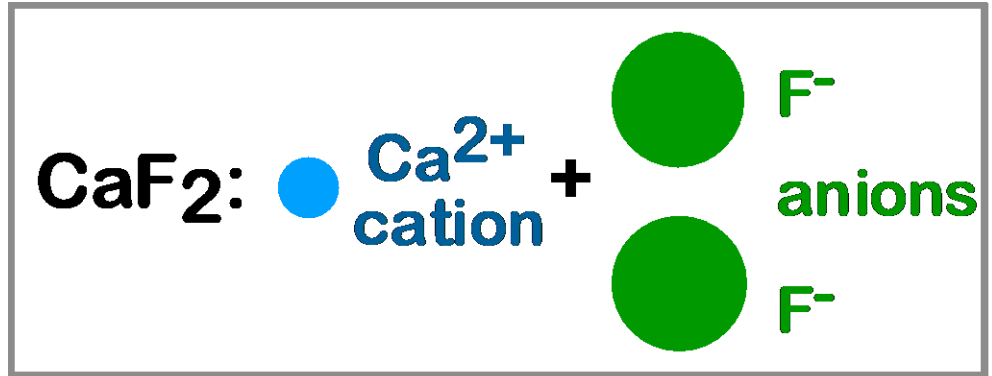
Result: theoretical $\rho_{\text{Cu}} = 8.89 \text{ g/cm}^3$

Compare to actual: $\rho_{\text{Cu}} = 8.94 \text{ g/cm}^3$

IONIC BONDING & STRUCTURE

- **Charge Neutrality:**

--Net charge in the structure should be zero.



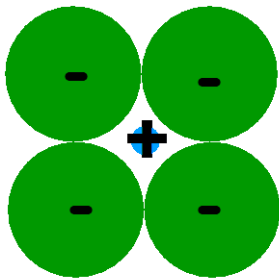
--General form:



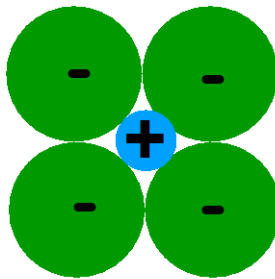
m, p determined by charge neutrality

- **Stable structures:**

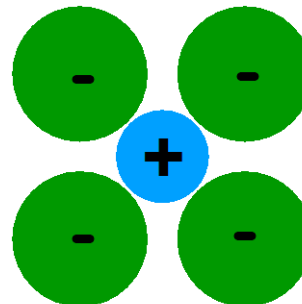
--maximize the # of nearest oppositely charged neighbors.



unstable



stable



stable

Adapted from Fig. 12.1,
Callister 6e.

COORDINATION # AND IONIC RADII

- Coordination # increases with

Issue: How many anions can you arrange around a cation?

r_{cation}
 r_{anion}

$\frac{r_{\text{cation}}}{r_{\text{anion}}}$ Coord #

< .155

2

.155-.225

3

.225-.414

4

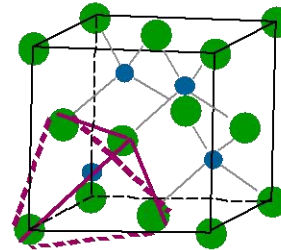
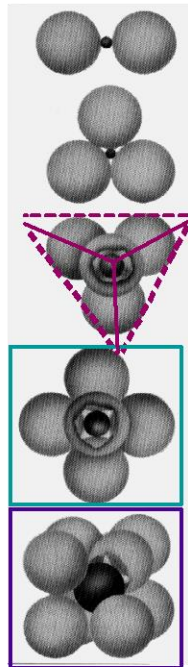
.414-.732

6

.732-1.0

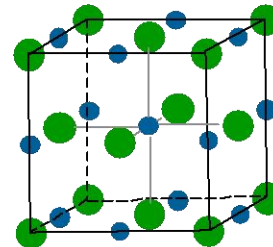
8

Adapted from Table 12.2, Callister 6e.



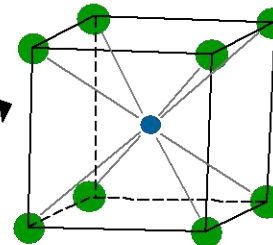
ZnS
(zincblende)

Adapted from Fig. 12.4, Callister 6e.



NaCl
(sodium chloride)

Adapted from Fig. 12.2, Callister 6e.



CsCl
(cesium chloride)

Adapted from Fig. 12.3, Callister 6e.

EX: PREDICTING STRUCTURE OF FeO

- On the basis of ionic radii, what crystal structure would you predict for FeO?

Cation Ionic radius (nm)

Al³⁺ 0.053

Fe²⁺ 0.077

Fe³⁺ 0.069

Ca²⁺ 0.100

Anion

O²⁻ 0.140

Cl⁻ 0.181

F⁻ 0.133

- Answer:

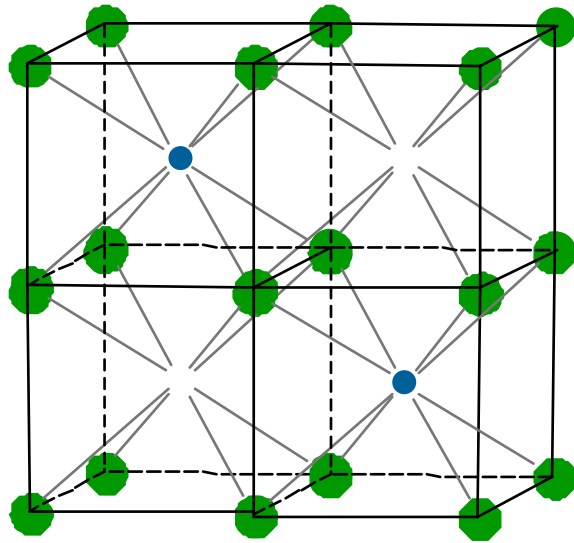
$$\frac{r_{\text{cation}}}{r_{\text{anion}}} = \frac{0.077}{0.140} = 0.550$$

based on this ratio,
--coord # = 6
--structure = NaCl

Data from Table 12.3,
Callister 6e.

A_mX_p STRUCTURES

- Consider CaF_2 : $\frac{r_{\text{cation}}}{r_{\text{anion}}} = \frac{0.100}{0.133} \approx 0.8$
- Based on this ratio, coord # = 8 and structure = CsCl.
- Result: CsCl structure w/only half the **cation** sites occupied.

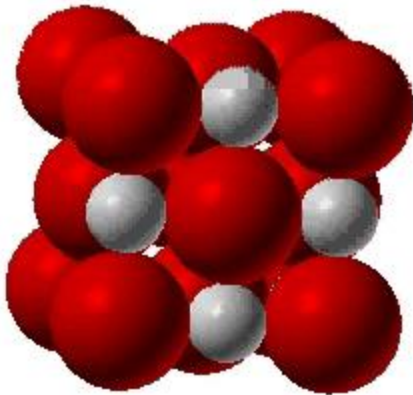


- Only half the **cation** sites are occupied since $\# \text{Ca}^{2+} \text{ ions} = 1/2 \# \text{F}^- \text{ ions}$.

Adapted from Fig. 12.5,
Callister 6e.

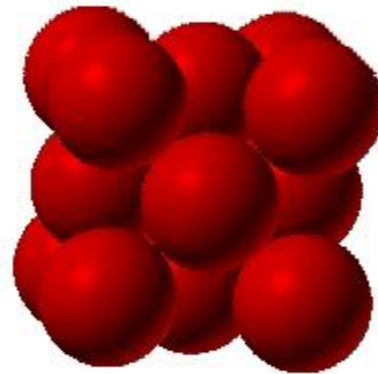
STRUCTURE OF COMPOUNDS: NaCl

- Compounds: Often have similar close-packed structures.
- Structure of NaCl
 - Close-packed directions
--along cube edges.



Click on image to animate

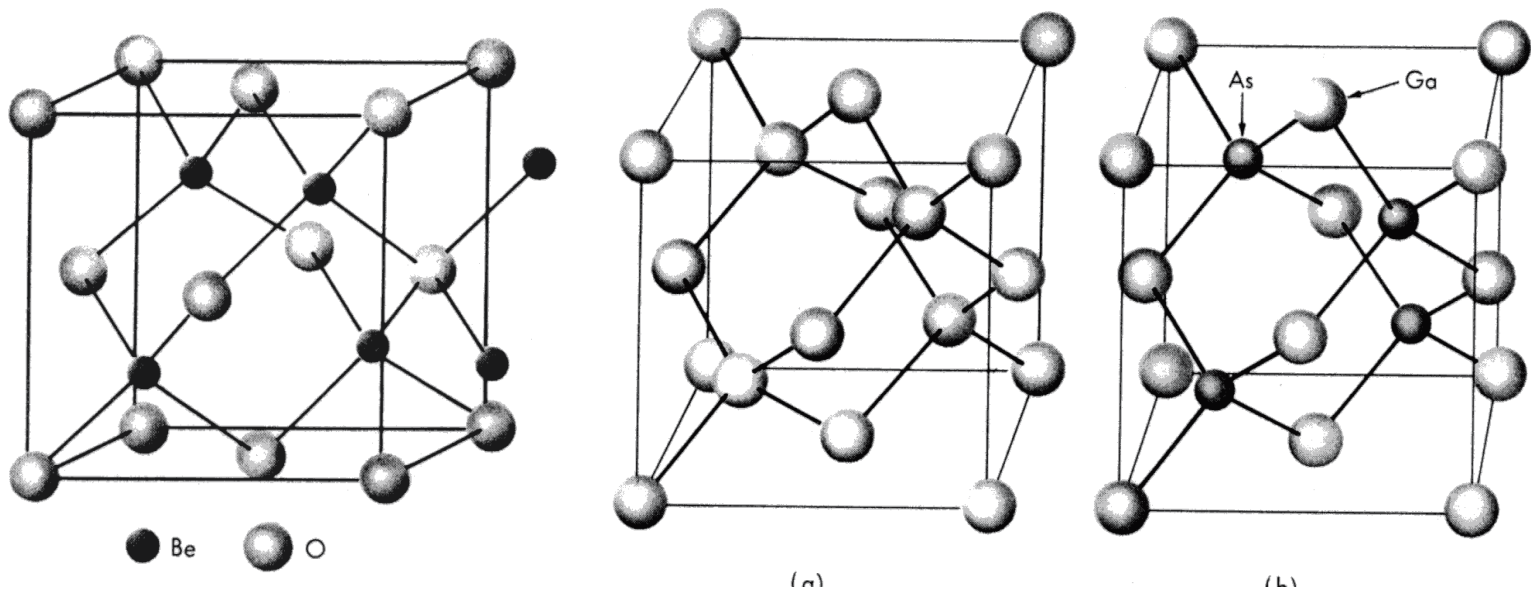
(Courtesy P.M. Anderson)



Click on image to animate

(Courtesy P.M. Anderson)

Diamond, BeO and GaAs are examples of FCC structures with two atoms per lattice point



SUMMARY

- Atoms may assemble into **crystalline** or **amorphous** structures.
- We can predict the **density** of a material, provided we know the **atomic weight**, **atomic radius**, and **crystal geometry** (e.g., FCC, BCC, HCP).
- Material properties generally vary with single crystal orientation (i.e., they are **anisotropic**), but properties are generally non-directional (i.e., they are **isotropic**) in polycrystals with randomly oriented grains.

Miller Indices

- The faces of the crystals both when they grow and when they are formed by cleavage
- Tend to be parallel either to the sides of the unit cell or to planes in the crystals.
- It is useful to be able to refer to both crystal faces and to the planes → by Miller indices.

Miller indices

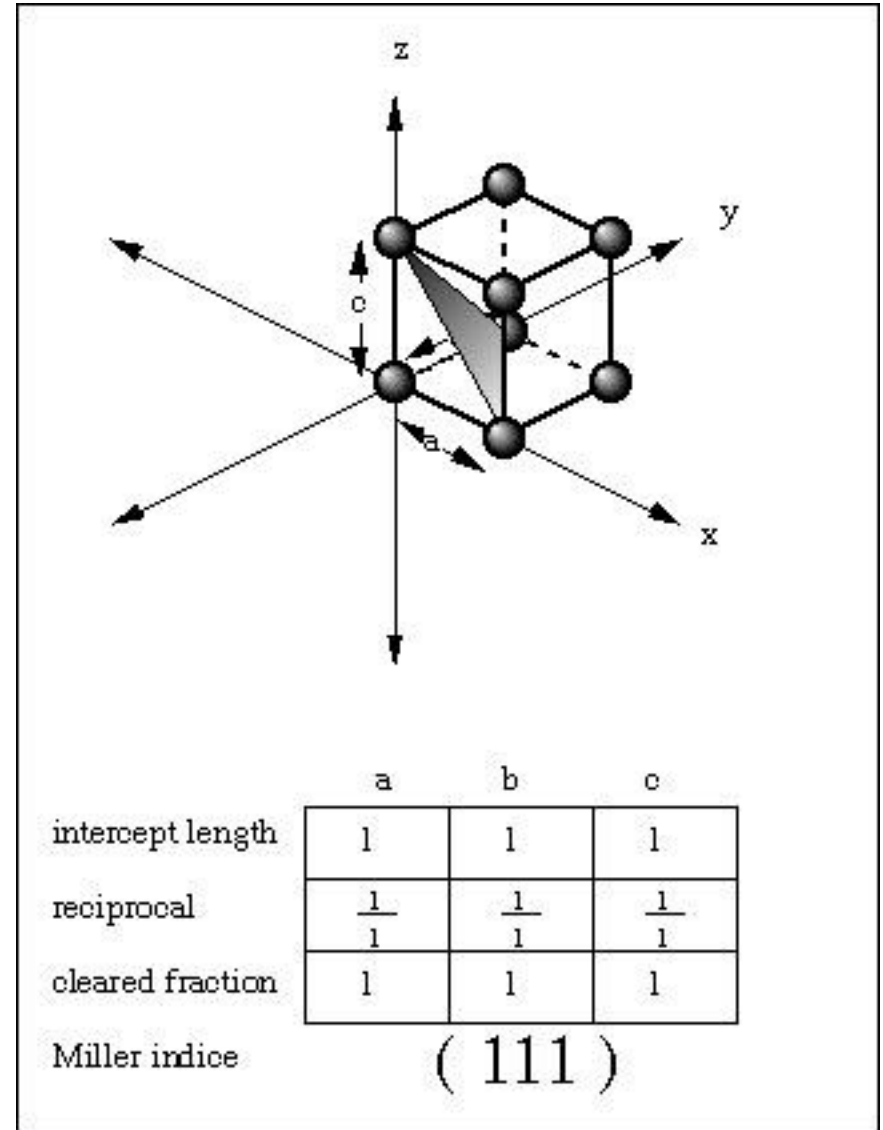
- A Miller index is a series of coprime integers that are inversely proportional to the intercepts of the crystal face or crystallographic planes with the edges of the unit cell.
- It describes the orientation of a plane in the 3-D lattice with respect to the axes.
- The general form of the Miller index is (h, k, l) where h , k , and l are integers related to the unit cell along the a , b , c crystal axes.

Miller Indices

Rules for determining Miller Indices:

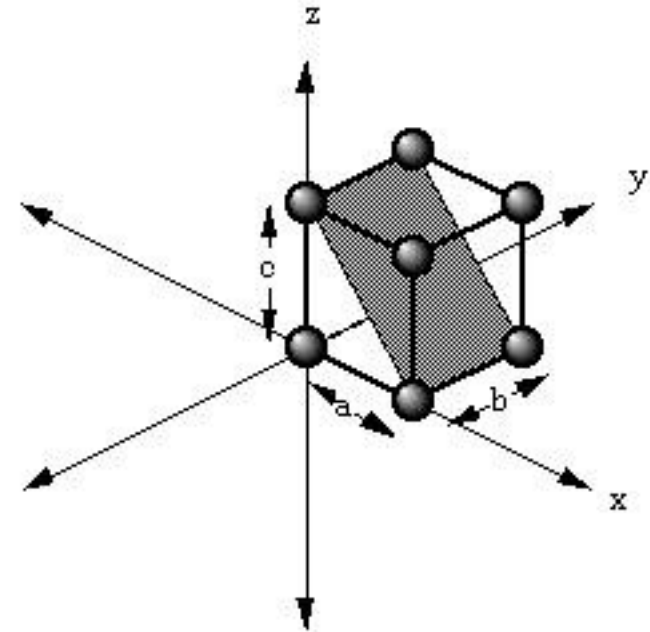
1. Determine the intercepts of the face along the crystallographic axes, *in terms of unit cell dimensions*.
2. Take the reciprocals
3. Clear fractions
4. Reduce to lowest terms

An example of the (111) plane ($h=1$, $k=1$, $l=1$) is shown on the right.



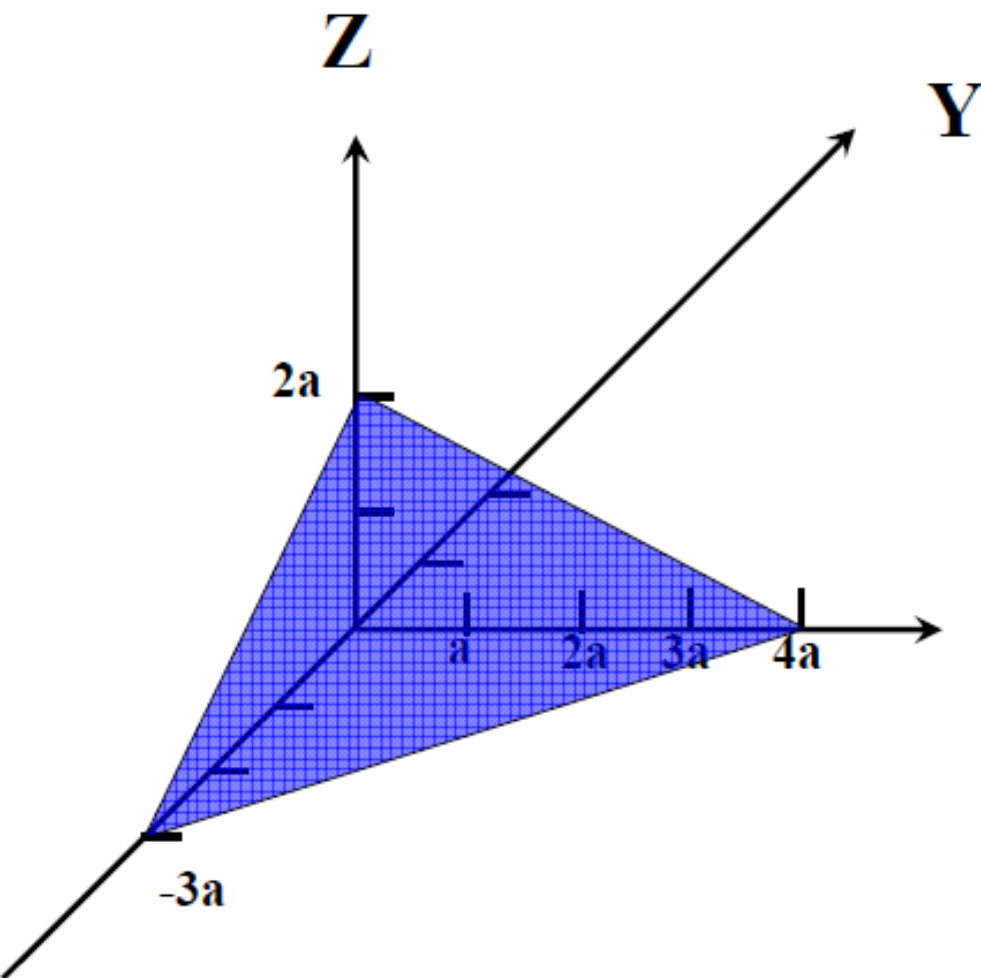
Another example:

In this case the plane intercepts the a axis at one unit length and also the c axis at one unit length. The plane however, never intersects the b axis. In other words, it can be said that the intercept to the b axis is infinity. The intercepts are then designated as 1,infinity,1. The reciprocals are then $\frac{1}{1}$, $\frac{1}{\infty}$, $\frac{1}{1}$. Knowing $\frac{1}{\infty} = 0$ then the indices become (101).



	a	b	c
intercept length	1	∞	1
reciprocal	$\frac{1}{1}$	$\frac{1}{\infty}$	$\frac{1}{1}$
cleared fraction	1	0	1
Miller indice	(101)		

Crystalline Planes and Miller Indices



Identify Intercepts in x,y,z
order = $4a, -3a, 2a$

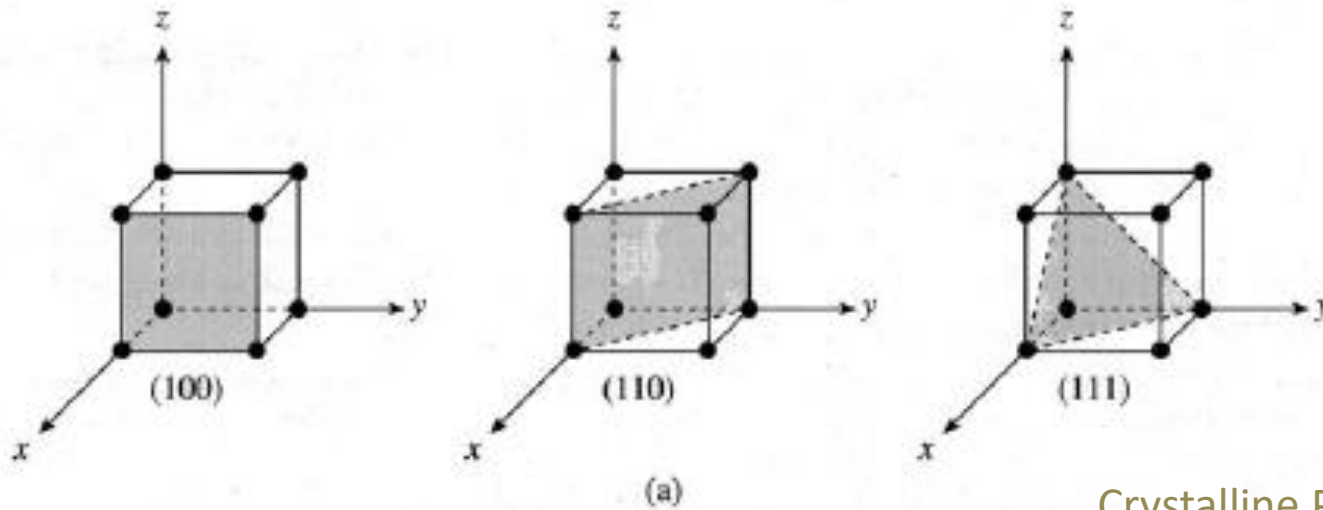
Divide by unit cell length in each
direction x,y,z order = $4, -3, 2$

Invert the values = $1/4, -1/3, 1/2$

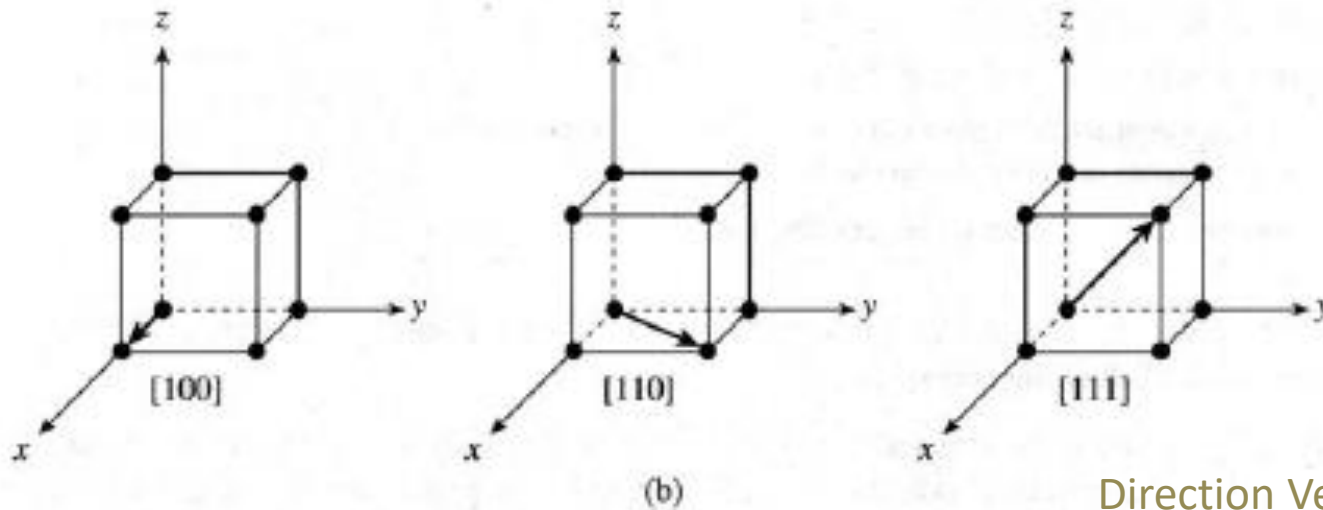
X Multiply by a number (12 in this
example) to give smallest whole
number set = $3, -4, 6$

Place any minus signs over their
index and place set in parenthesis
= $(\bar{3}46)$

Crystalline Planes and Miller Indices: Planes and directions



Crystalline Planes



Direction Vectors

Where does a protein crystallographer see the Miller indices?

- Common crystal faces are parallel to lattice planes

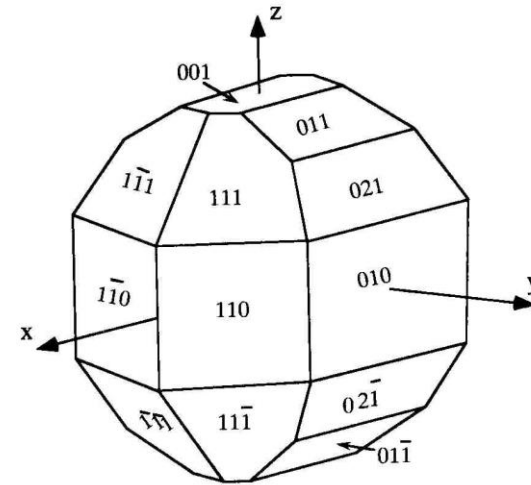
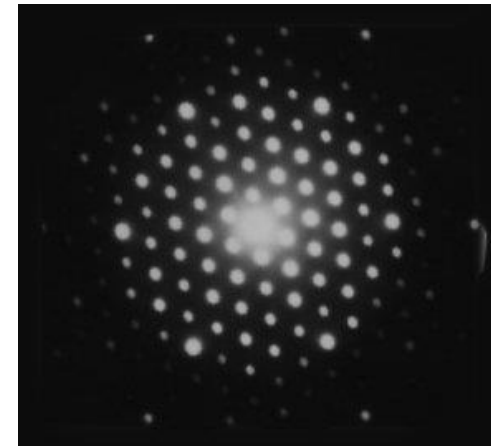


Figure 3.7. A crystal showing several faces.

- Each diffraction spot can be regarded as a X-ray beam reflected from a lattice plane, and therefore has a unique Miller index.

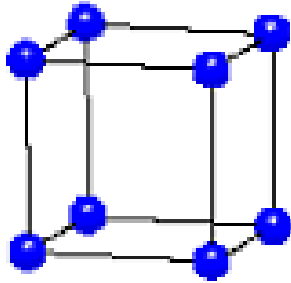


How many different lattice types exist for crystalline material?

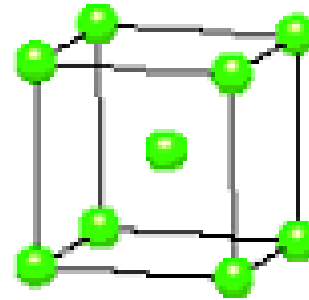
There are four types of lattices:

- 1) Primitive cubic
- 2) Body centered cubic
- 3) Face centered cubic
- 4) C-centered cubic

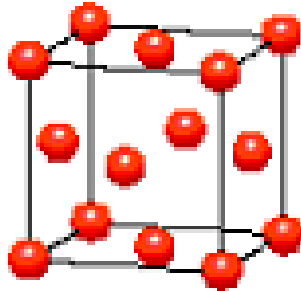
LATTICE TYPES



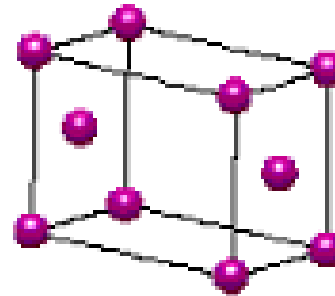
Primitive (**P**)



Body Centered (**I**)



Face Centered (**F**)



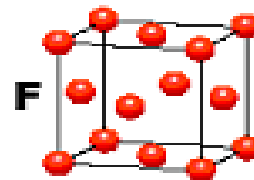
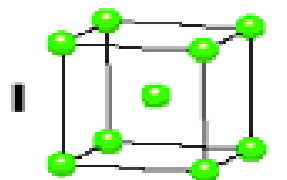
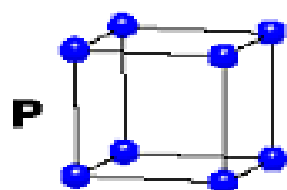
C-Centered (**C**)

BRAVAIS LATTICES

CUBIC

$$a = b = c$$

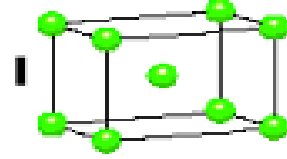
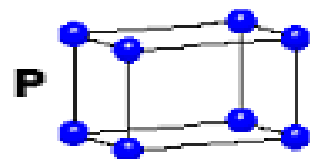
$$\alpha = \beta = \gamma = 90^\circ$$



TETRAGONAL

$$a = b \neq c$$

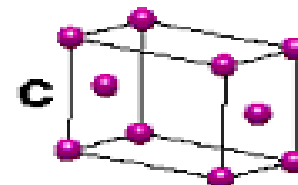
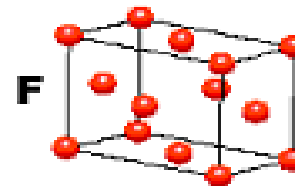
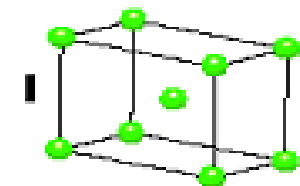
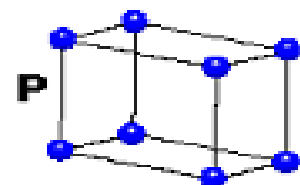
$$\alpha = \beta = \gamma = 90^\circ$$



ORTHORHOMBIC

$$a \neq b \neq c$$

$$\alpha = \beta = \gamma = 90^\circ$$

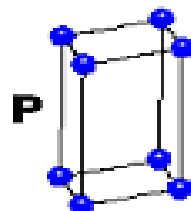


HEXAGONAL

$$a = b \neq c$$

$$\alpha = \beta = 90^\circ$$

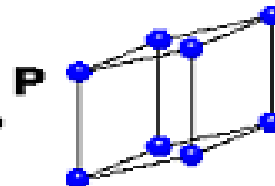
$$\gamma = 120^\circ$$



TRIGONAL

$$a = b = c$$

$$\alpha = \beta = \gamma \neq 90^\circ$$

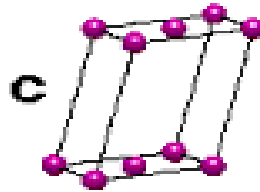
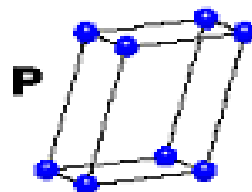


MONOCLINIC

$$a \neq b \neq c$$

$$\alpha = \gamma = 90^\circ$$

$$\beta \neq 120^\circ$$



TRICLINIC

$$a \neq b \neq c$$

$$\alpha \neq \beta \neq \gamma \neq 90^\circ$$

